Pulsar Timing Arrays and Gravitational Waves

AJ BDAY

Marc Kamionkowski · Johns Hopkins University · 30 June 2026

Andrew → Brian Keating→BICEP/KEck

PHYSICAL REVIEW D, VOLUME 61, 083501

Polarization pursuers' guide

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(Received 17 September 1999; published 1 March 2000)

We calculate the detectability of the polarization of the cosmic microwave background (CMB) as a function of the sky coverage, angular resolution, and instrumental sensitivity for a hypothetical experiment. We consider the gradient component of the polarization from density perturbations (scalar modes) and the curl component from gravitational waves (tensor modes). We show that the amplitude (and thus the detectability) of the polarization from density perturbations is roughly the same in any model as long as the model fits the big-bang-nucleosynthesis (BBN) baryon density and degree-scale anisotropy measurements. The degree-scale polarization is smaller (and accordingly more difficult to detect) if the baryon density is higher. We show that the sensitivity to the polarization from density perturbations and gravitational waves is improved (by a factor of 30) in a fixed-time experiment with a deeper survey of a smaller region of sky.

The angular three-point correlation function in the quasilinear regime

Ari Buchalter, Marc Kamionkowski, Andrew H. Jaffe (Mar, 1999)

Published in: *Astrophys.J.* 530 (2000) 36 • e-Print: astro-ph/9903486 [astro-ph]

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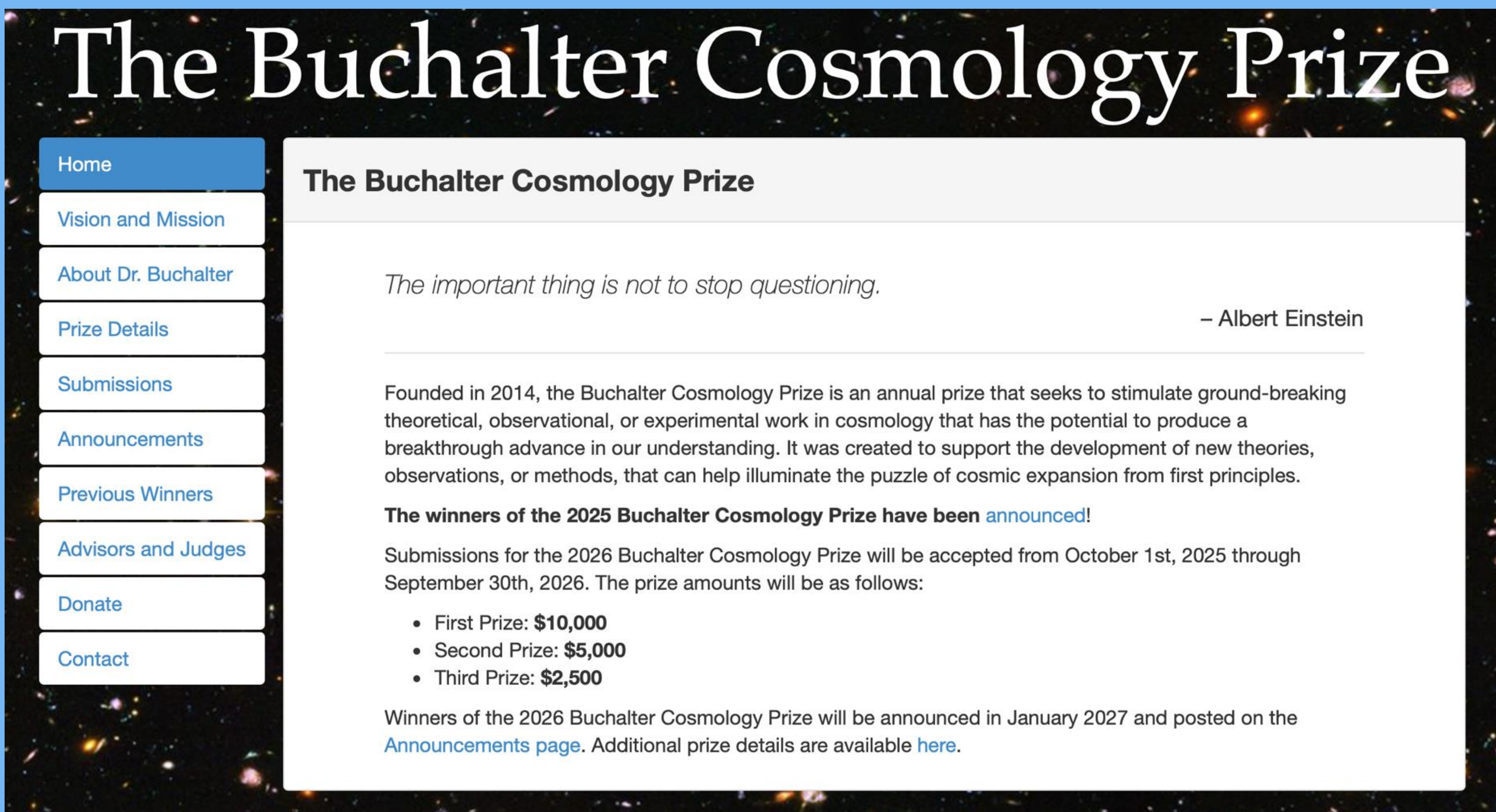

The Buchalter Cosmology Prize

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The Buchalter Cosmology Prize

The important thing is not to stop questioning.

– Albert Einstein

Founded in 2014, the Buchalter Cosmology Prize is an annual prize that seeks to stimulate ground-breaking theoretical, observational, or experimental work in cosmology that has the potential to produce a breakthrough advance in our understanding. It was created to support the development of new theories, observations, or methods, that can help illuminate the puzzle of cosmic expansion from first principles.

The winners of the 2025 Buchalter Cosmology Prize have been announced!

Submissions for the 2026 Buchalter Cosmology Prize will be accepted from October 1st, 2025 through September 30th, 2026. The prize amounts will be as follows:

- First Prize: **$10,000**
- Second Prize: **$5,000**
- Third Prize: **$2,500**

Winners of the 2026 Buchalter Cosmology Prize will be announced in January 2027 and posted on the Announcements page. Additional prize details are available here.

PHYSICAL REVIEW D, VOLUME 58, 043001

Calculation of the Ostriker-Vishniac effect in cold dark matter models

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Random Universe → Random Crap

Assembling these results, in a flat universe, the Ostriker-Vishniac effect produces a power spectrum,

$$P_p(\kappa)=\frac{1}{16\pi^2}\int_0^{\eta_0}\frac{g^2(w)}{w^2}[a(w)]^2\left(\frac{\dot{D}D}{D_0^2}\right)^2 S(\kappa/w)dw, \tag{22}$$

with the Vishniac power spectrum $S(k)$ given by Eq. (21). In an open or closed universe, one makes the replacement Eq. (A10) in the argument of $S(\kappa/w)$ and in the two factors of w in the denominator of the integrand the numbers for the open models in the table and the open-model curves need to be fixed.

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The Cosmos Is Thrumming With Gravitational Waves, Astronomers Find

Radio telescopes around the world picked up a telltale hum reverberating across the cosmos, most likely from supermassive black holes merging in the early universe.

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The Very Large Array on the Plains of San Agustin, N.M., one of three radio telescopes that worked with a global consortium to detect the timing of pulsars. NRAO/AUI/NSF

By Katrina Miller

Katrina Miller, a science reporter, recently earned a Ph.D. in particle physics from the University of Chicago.

June 28, 2023

Jaffe & Backer (2003) got the ball rolling

THE ASTROPHYSICAL JOURNAL, 583:616–631, 2003 February 1

GRAVITATIONAL WAVES PROBE THE COALESCENCE RATE OF MASSIVE BLACK HOLE BINARIES

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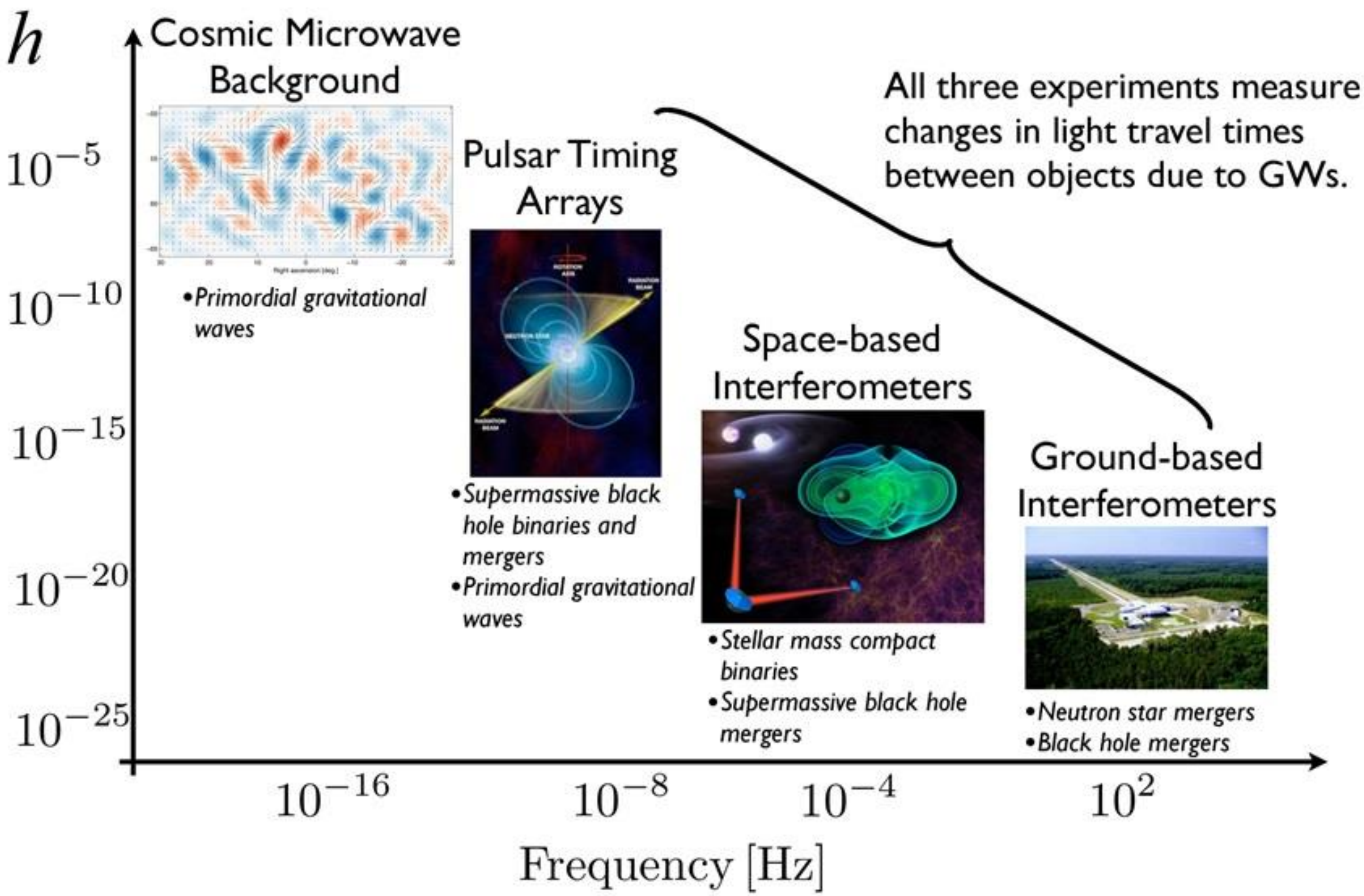
Received 2002 February 27; accepted 2002 October 8

Connected black-hole demographics to the possibility to detect nHz GWs with PTAs and thus got the ball rolling

The spectrum of gravitational wave astronomy



Monthly Notices of the ROYAL ASTRONOMICAL SOCIETY

MNRAS **487,** 562–579 (2019) doi:10.1093/mnras/stz1022
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On the amplitude and Stokes parameters of a stochastic gravitational-wave background

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Pulsar-timing arrays, astrometry, and gravitational waves

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The search for statistical anisotropy in the gravitational-wave background with pulsar timing arrays

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September 2019

$$z(\hat{n}) \to z_{\ell m}$$

$$\mathrm{HD}(\Theta) \to C_\ell \quad \left(\propto (\ell-2)!/(\ell+2)!\right)$$

$$\langle z_{\ell m} z^*_{\ell' m'} \rangle = C_\ell \delta_{\ell\ell'} \delta_{mm'} + \sum_{L=1}^{\infty} \sum_{M=-L}^{L} (-1)^{m'} \langle \ell\, m\, \ell', -m' | LM \rangle A^{LM}_{\ell\ell'},$$

Anisotropy parameterized by BiPoSHs *(thanks, Tarun!)*

$$\left\langle h_s(\vec{k}) h^*_{s'}(\vec{k}') \right\rangle = \frac{1}{4}\delta_{ss'}(2\pi)^3 \delta_D(\vec{k} - \vec{k}') P_h(k) \times \left[1 + \sum_{L>0} \sum_{M=-L}^{L} g_{LM} Y_{LM}(\hat{k})\right],$$

$$A^{LM}_{\ell\ell'} = (-1)^{\ell-\ell'} (4\pi)^{-1/2} g_{LM} z_\ell z_{\ell'} H^L_{\ell\ell'} I,$$

Then minimum-variance estimators are

$$\widehat{g_{LM}} = \frac{\sum_{\ell\ell'} (\widehat{g_{LM}})_{\ell\ell'} (\Delta g_{LM})^{-2}_{\ell\ell'}}{\sum_{\ell\ell'} (\Delta g_{LM})^{-2}_{\ell\ell'}}.$$

with variances

$$(\Delta g_{LM})^{-2} = \frac{27}{16\pi^3} \sum_{\ell\ell'} \frac{\left[H^L_{\ell\ell'} z_\ell z_{\ell'} (\mathrm{SNR}) N^{zz}\right]^2}{(C_\ell + N^{zz})(C_{\ell'} + N^{zz})}.$$

- SNR is $\propto \left[\frac{(l-2)!}{(l+2)!}\right]^2$ --- *overwhelmingly* in quadrupole

$$(\mathrm{SNR})^2 = \sum_{\ell=2}^{\ell_{\max}} \frac{(2\ell+1)}{2} \left(\frac{C_\ell}{N^{zz}}\right)^2$$

- → Will need *very* high SNR to characterize signal after its detection

GWs and astrometry (Qin, Boddy, MK, Dai, 2019)

$$(\delta n)_a(\hat{\boldsymbol{n}}) = \sum_{lm} \left[E_{\ell m} Y^E_{(\ell m)a}(\hat{\boldsymbol{n}}) + B_{\ell m} Y^B_{(\ell m)a}(\hat{\boldsymbol{n}}) \right], \quad (3)$$

in terms of spherical-harmonic coefficients,

$$E_{\ell m} = \int d\hat{\boldsymbol{n}}\, (\delta n)^a(\hat{\boldsymbol{n}}) Y^E_{(\ell m)a}(\hat{\boldsymbol{n}}),$$

$$B_{\ell m} = \int d\hat{\boldsymbol{n}}\, (\delta n)^a(\hat{\boldsymbol{n}}) Y^B_{(\ell m)a}(\hat{\boldsymbol{n}}). \quad (4)$$

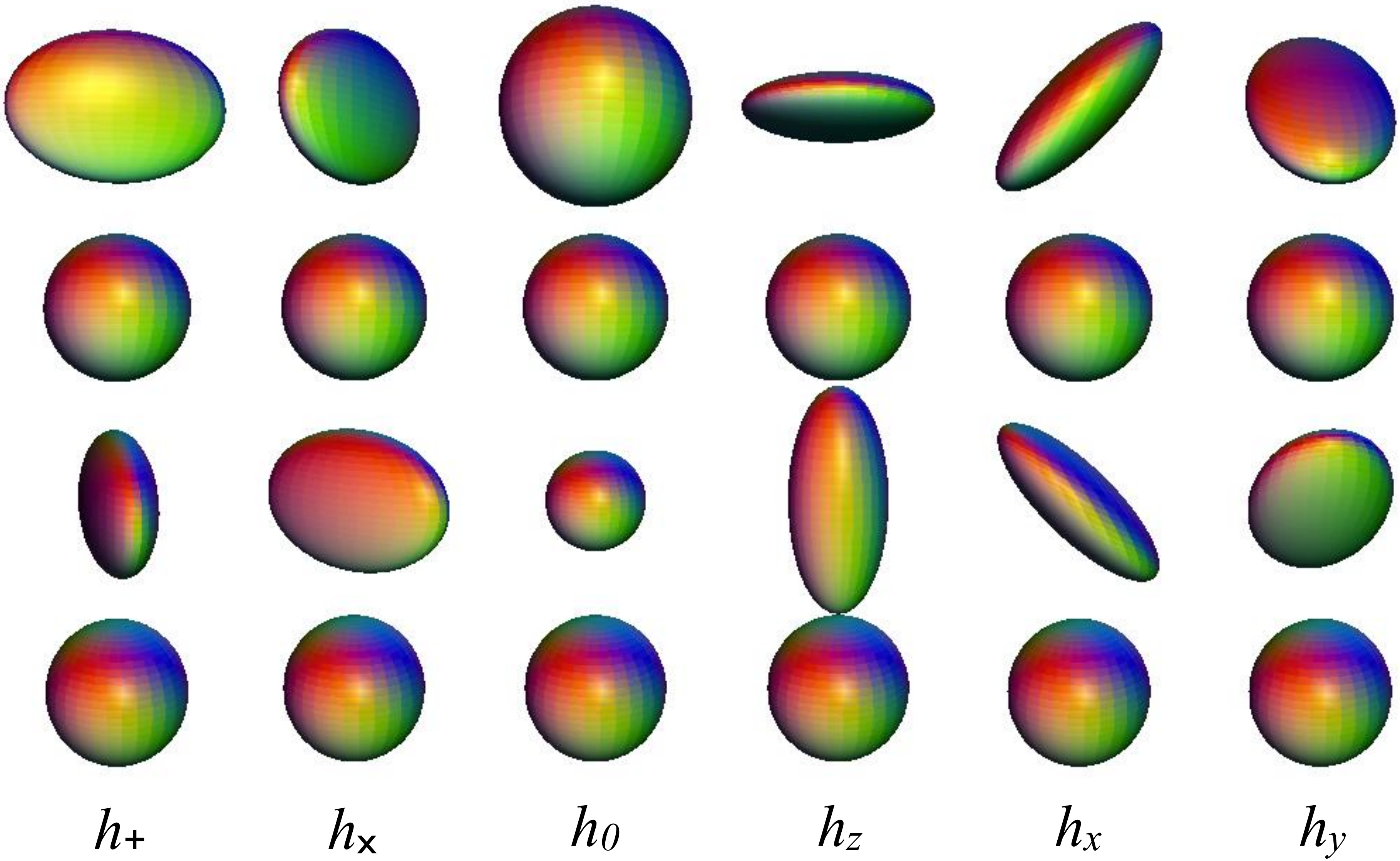
$$C_\ell^{XX'} = \frac{1}{2\ell+1} \sum_m X_{\ell m} \left(X'_{\ell m}\right)^* ,$$

$$C_\ell^{zz,\alpha} = \sum_k \left(C_\ell^{zz}\right)_{k\ell m} = \frac{4}{\pi} \left|F_\ell^{z,\alpha}\right|^2 \int k^2\, dk\, P_h(k) \left[W(k)\right]^2$$

X	$F_\ell^{E,X}$	$F_\ell^{B,X}$	$F_\ell^{z,X}$
ST	$\frac{i}{6}\delta_{\ell 1}$	0	$-\frac{1}{2\sqrt{2}}\left(\delta_{\ell 0}+\frac{i}{3}\delta_{\ell 1}\right)$
SL	$-\frac{i}{3\sqrt{2}}\delta_{\ell 1}+\frac{i^\ell}{2\sqrt{\ell(\ell+1)}}$	0	$-\frac{i^\ell}{4}\ln(kr_s)$
VE	$\frac{2i}{3\sqrt{2}}\delta_{\ell 1}-\frac{i^\ell}{\sqrt{2}\ell(\ell+1)}$	0	$-\frac{i}{3}\delta_{\ell 1}+\frac{i^\ell}{\sqrt{2\ell(\ell+1)}}$
VB	0	$\frac{i}{3\sqrt{2}}\delta_{\ell 1}-\frac{i^\ell}{\sqrt{2}\ell(\ell+1)}$	0
TE	$-i^\ell\frac{N_\ell^{-1}}{\sqrt{\ell(\ell+1)}}$	0	$\frac{i^\ell}{2}N_\ell^{-1}$
TB	0	$-i^\ell\frac{N_\ell^{-1}}{\sqrt{\ell(\ell+1)}}$	0

All six auto- and cross-correlations for PTA/astrometry observables for each of the 6 possible GW polarizations (Qin et al. 2019). Reproduces and extends prior results in configuration space

GW polarizations (from Jeong & MK, 2012)



In TAM-wave formalism,

$$\{+, \times, 0, L, x, y\} \rightarrow \{TE, TB, ST, SL, VE, VB\}$$

If GW v<1, generalizes to (Qin,Boddy,MK 2021)

α	$F_\ell^{E,\alpha}$	$F_\ell^{B,\alpha}$	$F_\ell^{z,\alpha}$
ST	$\frac{i}{6v}\delta_{\ell 1} - \frac{1-v^2}{2v^2}\sqrt{\frac{\ell(\ell+1)}{2}} I_\ell^{(1)}(v)$	0	$-\frac{1}{2\sqrt{2}}\left[\frac{1}{v^2}\delta_{\ell 0} + \frac{i}{3v}\delta_{\ell 1} + i\frac{1-v^2}{v^3} I_\ell^{(0)}(v)\right]$
SL	$-\frac{i}{3\sqrt{2}v}\delta_{\ell 1} + \frac{1}{2v^2}\sqrt{\ell(\ell+1)} I_\ell^{(1)}(v)$	0	$\frac{1}{2}\left(\frac{1}{v^2}\delta_{\ell 0} + \frac{i}{3v}\delta_{\ell 1}\right) + \frac{i}{2v^3} I_\ell^{(0)}(v)$
VE	$\frac{2i}{3\sqrt{2}v}\delta_{\ell 1} - \frac{1}{\sqrt{2}v^2} I_\ell^{(1)}(v) + \frac{i(1-v^2)}{\sqrt{2}v^3} I_\ell^{(0)}(v)$	0	$-\frac{i}{3v}\delta_{\ell 1} + \frac{1}{v^2}\sqrt{\frac{\ell(\ell+1)}{2}} I_\ell^{(1)}(v)$
VB	0	$\frac{i}{3\sqrt{2}}\delta_{\ell 1} - \frac{1}{\sqrt{2}v} I_\ell^{(1)}(v)$	0
TE	$-\frac{N_\ell}{\sqrt{\ell(\ell+1)}}\left[\frac{i}{v} I_\ell^{(2)}(v) + \frac{1-v^2}{2v^2} I_\ell^{(1)}(v)\right]$	0	$\frac{i}{2v} N_\ell I_\ell^{(2)}(v)$
TB	0	$-\frac{iN_\ell}{\sqrt{\ell(\ell+1)}} I_\ell^{(2)}(v)$	0

But what if GW background is also linearly/circularly polarized?

ORFs for intensity & circular/linear-polarization anisotropies

(Anil Kumar and MK, 2024)

$$\langle \tilde{h}_p^*(f,\hat{\Omega})\tilde{h}_{p'}(f',\hat{\Omega}')\rangle = \delta_D(f-f')\delta_D^2(\hat{\Omega},\hat{\Omega}')\mathcal{P}_{p,p'}(f,\hat{\Omega}), \tag{1}$$

$$\mathcal{P}_{p,p'}(f,\hat{\Omega}) = \begin{pmatrix} I(f,\hat{\Omega})+Q(f,\hat{\Omega}) & U(f,\hat{\Omega})-iV(f,\hat{\Omega}) \\ U(f,\hat{\Omega})+iV(f,\hat{\Omega}) & I(f,\hat{\Omega})-Q(f,\hat{\Omega}) \end{pmatrix}, \tag{2}$$

$$\begin{aligned}
I(f,\hat{\Omega}) &= \frac{1}{2}\langle|\tilde{h}_+|^2+|\tilde{h}_\times|^2\rangle,\\
Q(f,\hat{\Omega}) &= \frac{1}{2}\langle|\tilde{h}_+|^2-|\tilde{h}_\times|^2\rangle,\\
U(f,\hat{\Omega}) &= \mathrm{Re}\langle\tilde{h}_+^*\tilde{h}_\times\rangle = \frac{1}{2}\langle\tilde{h}_+^*\tilde{h}_\times+\tilde{h}_\times^*\tilde{h}_+\rangle,\\
V(f,\hat{\Omega}) &= \mathrm{Im}\langle\tilde{h}_+^*\tilde{h}_\times\rangle = \frac{1}{2i}\langle\tilde{h}_+^*\tilde{h}_\times-\tilde{h}_\times^*\tilde{h}_+\rangle,
\end{aligned}$$

$$I(f,\hat{\Omega}) = I_0\sum_{L=0}^{\infty}\sum_{M=-L}^{L} c_{LM}^{I} Y_{LM}(\hat{\Omega}),$$

$$V(f,\hat{\Omega}) = I_0\sum_{L=0}^{\infty}\sum_{M=-L}^{L} c_{LM}^{V} Y_{LM}(\hat{\Omega}),$$

$$P_\pm(f,\hat{\Omega}) = I_0\sum_{L=4}^{\infty}\sum_{M=-L}^{L} c_{LM\pm4}^{\pm} Y_{LM}(\hat{\Omega}),$$

$$P_\pm = Q \pm iU$$

$$c_{LM}^{E} = \frac{1}{2}\left(c_{LM}^{+}+c_{LM}^{-}\right), \quad c_{LM}^{B} = -\frac{i}{2}\left(c_{LM}^{+}-c_{LM}^{-}\right). \tag{7}$$

$$\Gamma^X_{LM}(\hat{n}_a,\hat{n}_b) = (-1)^L\sqrt{\pi}\sum_{\ell\ell'} F^{L,X}_{\ell\ell'}\,\{Y_\ell(\hat{n}_a)\otimes Y_{\ell'}(\hat{n}_b)\}_{LM}\,.$$

$$\{Y_\ell(\hat{n}_a)\otimes Y_{\ell'}(\hat{n}_b)\}_{LM} = \sum_{mm'}\langle \ell m \ell' m' | LM\rangle Y_{\ell m}(\hat{n}_a) Y_{\ell' m'}(\hat{n}_b),$$

X	Spin-2	Spin-1
I	$2z^t_\ell z^t_{\ell'}\begin{pmatrix}\ell & \ell' & L\\ -2 & 2 & 0\end{pmatrix}\dfrac{X^L_{\ell\ell'}}{(1+\delta_{\ell\ell'})/2}$	$z^v_\ell z^v_{\ell'}\begin{pmatrix}\ell & \ell' & L\\ -1 & 1 & 0\end{pmatrix}\dfrac{X^L_{\ell\ell'}}{(1+\delta_{\ell\ell'})/2}$
V	$2z^t_\ell z^t_{\ell'}\begin{pmatrix}\ell & \ell' & L\\ -2 & 2 & 0\end{pmatrix}(1-X^L_{\ell\ell'})$	$z^v_\ell z^v_{\ell'}\begin{pmatrix}\ell & \ell' & L\\ -1 & 1 & 0\end{pmatrix}(1-X^L_{\ell\ell'})$
E	$2z^t_\ell z^t_{\ell'}\begin{pmatrix}\ell & \ell' & L\\ 2 & 2 & -4\end{pmatrix}\dfrac{X^L_{\ell\ell'}}{(1+\delta_{\ell\ell'})/2}$	$z^v_\ell z^v_{\ell'}\begin{pmatrix}\ell & \ell' & L\\ 1 & 1 & -2\end{pmatrix}\dfrac{X^L_{\ell\ell'}}{(1+\delta_{\ell\ell'})/2}$
B	$2z^t_\ell z^t_{\ell'}\begin{pmatrix}\ell & \ell' & L\\ 2 & 2 & -4\end{pmatrix}(1-X^L_{\ell\ell'})$	$z^v_\ell z^v_{\ell'}\begin{pmatrix}\ell & \ell' & L\\ 1 & 1 & -2\end{pmatrix}(1-X^L_{\ell\ell'})$

Hotinli, MK, Jaffe 2019

Belgacem & MK 2020

Chu, Liu, Ng 2021
Chu & Ng 2022
Anil Kumar et al. 2024

Anil Kumar
& MK, 2024

Can generalize to astrometry

Inomata, MK, Toral, Taylor, 2024

$$\Gamma^{p,X,ST}_{(LM)ij}(\hat{n}_a,\hat{n}_b) = (-1)^L\sqrt{\pi}\sum_{\ell=\ell_{\min}}^{\ell_{\max}}\sum_{\ell'=\ell_{\min}}^{\ell_{\max}} F^{L,p,X,ST}_{\ell\ell'}\left\{Y^S_{(\ell)i}(\hat{n}_a)\otimes Y^T_{(\ell')j}(\hat{n}_b)\right\}_{LM}, \tag{52}$$

Chiral gravitational waves
(Lue, Wang, MK, 1999)

- Chern-Simons gravity during inflation

$$\phi R\tilde{R}$$

- May lead to right/left asymmetry ("amplitude biregringence")

GW (angle-independent) chirality and PTAs/astrometry

- Cannot be detected with PTAs
- With astrometry, CAN be detected with TAM TE/TB or z/TB correlations

Summary

- Detection/characterization of SGWB is important but difficult
- Better prospects to detect it (soon) with PTAs/astrometry